

KiCad StepUp tools cheat sheet

<https://github.com/easyw/kicadStepUpMod>

1) What StepUp tools are for?

KiCad StepUp tools are a [FreeCAD Macro](#) and a [FreeCAD WorkBench](#) to help in **Mechanical Collaboration** between KiCad EDA and a Mechanical CAD.

With StepUp it is possible to:

- load kicad board and parts in FreeCAD and export it to STEP (or IGES) for a full ECAD MCAD collaboration
- load *kicad_mod* footprint in FreeCAD to easy and precisely align the mechanical model to kicad footprint
- convert the STEP 3D model of parts, board, enclosure to VRML with Materials properties for the best use in kicad
- check interference and collisions for enclosure and footprint design
- design a new pcb Edge with FreeCAD Sketcher and PUSH it to an existing kicad_pcb Board
- PULL a pcb Edge from a kicad_pcb Board, edit it in FC Sketcher and PUSH it back to kicad
- PUSH & PULL 3D models positions between FreeCAD and KiCAD
- ECAD / MCAD Collaboration and Synchronization (compressed 'stpZ' format allowed)
- footprint generation even for complex pads and shapes
- generate Blender compatible VRML files

2) Requirements

KiCad StepUp tools need with the following requirements:

- **KiCad Stable Release >= 4.0** or kicad **Nightly Development Builds**
- **FreeCAD** stable release >= **0.18** (FC daily is supported too)
- a library of STEP 3D models now available as default from [KiCad/packages3D](#)

3) How to install StepUp tools

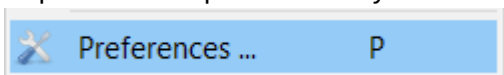
KiCad StepUp tools can be installed as a **FreeCAD Macro** but it is strongly suggested to install StepUp as a **FreeCAD WorkBench**.

Since KiCad StepUp tools have been added to [FC WorkBenches](#), so they can be installed through the FreeCAD [addons installer](#) or starting from FC version 0.17, through the **addons manager** in the FC Tools Menu. Then StepUp buttons will be available to be customized in FC Toolbars.

If KiCad StepUp tools are installed as a FC WorkBench, then **it will be possible to Open directly from the FC File Menu a *kicad_pcb* board file or a *kicad_mod* footprint file** and many useful features will be also available.

4) Configure StepUp tools

To use StepUp tools for converting a *kicad_pcb* Board to a mechanical STEP model you just need to **configure** your 3D prefix path(s) like your **KISYS3DMOD**. These settings are found the FreeCAD StepUp preferences page which only appear when using the **KiCadStepUp Workbench**. With that workbench activated, you can configure the paths via the preferences system of FreeCAD (Edit menu -> Preferences).



Just click the green icon:



Using **Snap** or **FlatPak** is requiring to 'bind mount' the 3D models folder
Some references at [GitHub Repo Issue](#)

5) Tips

Tips to use StepUp tools at its best

- never use a scale different from 1:1:1 in your 3D models
 - configure your [prefix3D] in the FreeCAD StepUp preference page to your KISYS3DMOD path
 - use STEP or STPZ or IGES or VRML or WRZ or mixed type of models in your board
 - use bounding boxes to reduce your STEP board file size if required
 - each 3D model is suggested to be a single object (union of parts or compound in FC)
- note:** compound may be slower than union, because it needs to re-create a compound after loading the model

6) Useful Video Tutorials

Here some links of StepUp tutorial:

- StepUp: [Align Parts to Kicad footprint](#)
- StepUp: [converting a KiCad board and Parts to STEP](#)
- StepUp: [PUSH & PULL a PCB Edge using FC Sketcher](#)
- StepUp: [PUSH&PULL 3D models between KiCAD & FreeCAD \(ECAD MCAD Synchronization\)](#)

There is also a video tutorial made by a user:

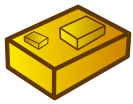
- StepUp: [Installing, Import 3D model, Exporting the Board](#)

Note: in the video the user is copying all demo files, when in fact it is better to install StepUp as a FreeCAD WorkBench.

7) Need Help?

KiCad info forum is a great resource:

<https://forum.kicad.info/search?q=step>



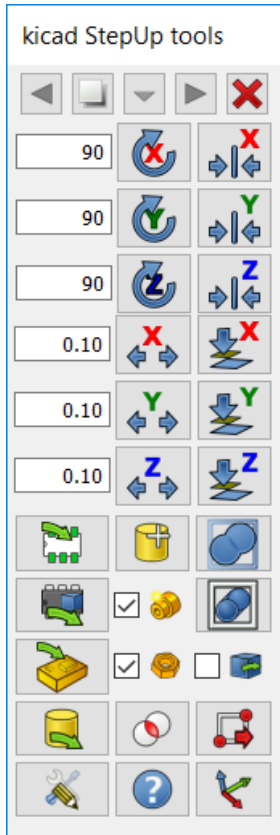
KiCad StepUp tools cheat sheet

<https://github.com/easyw/kicadStepUpMod>

The Main Gui

A brief recap on KiCad StepUp tools buttons.

Note: each button has a Tooltip



Load 'kicad_pcb' Board

Load a 'kicad_pcb' file into FreeCAD



Import 3D model to be Aligned

Import a 3D STEP model into FreeCAD



Load 'kicad_mod' Footprint

Load a 'kicad_mod' footprint into FreeCAD



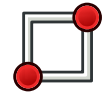
Export 3D model Aligned

Export a 3D STEP & VRML model back to KiCad



Export selected to STEP

Export selected objects or Board and Parts to hierarchical STEP file



Push & Pull PCB Edge

Read and Write pcb Edge from KiCad into FC Sketcher



Add Reference Axis

Add reference Axis to the FreeCAD design



Check Interferences and Collisions

Check Interference and Collisions in Board Design



Help

Mini Help inside StepUp tools



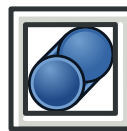
Preferences Config Page

showing the preferences Page



Make a Union

Make a Union of Parts



Make a Compound

Make a Compound of Parts

Option checkboxes



Materials properties

Adding Material to VRML when Exporting a 3D model



Virtual mechanical

Adding Virtual kicad Parts when Loading a 3D model of the PCB



export Board to STEP

Automatically export Board & Parts to STEP after Loading a 3D model of the PCB if checked

Useful Video Tutorials

Here some links of StepUp tutorial:

- StepUp: [Align Parts to Kicad footprint](#)
- StepUp: [converting a KiCad board and Parts to STEP](#)
- StepUp: [PUSH & PULL a PCB Edge using FC Sketcher](#)
- StepUp: [ECAD MCAD Synchronization & Collaboration](#)

There is also a video tutorial made by a user:

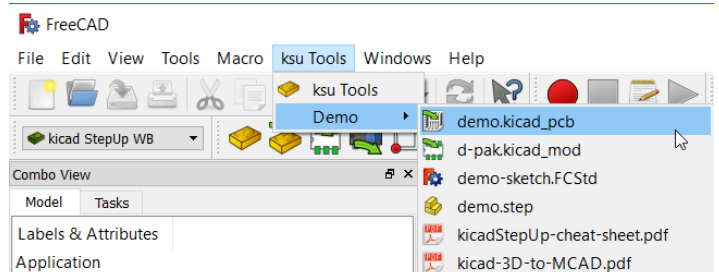
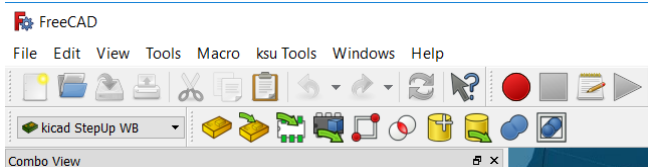
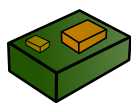
- StepUp: [Installing, Import 3D model, Exporting the Board](#)

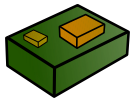
Note: in the video the user is copying all demo files, when in fact it is only needed *kicad-StepUp-tools.FCMacro* file.

The WorkBench

A screenshot on KiCad StepUp WB.

Demo and Manuals in the StepUp WB Menu





KiCad StepUp tools cheat sheet

<https://github.com/easyw/kicadStepUpMod>

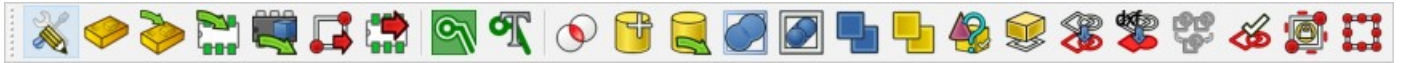
The WorkBench

A screenshot on KiCad StepUp WB.

Note: each button has a useful Tooltip

Demo and Manuals in the StepUp WB Menu

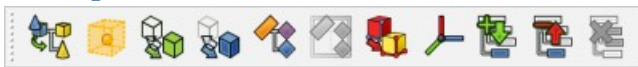
Main Tool bar



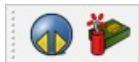
Push&Pull Tool bar



Helpers



Show tools



Useful Designing external workbenches



Two external workbenches:

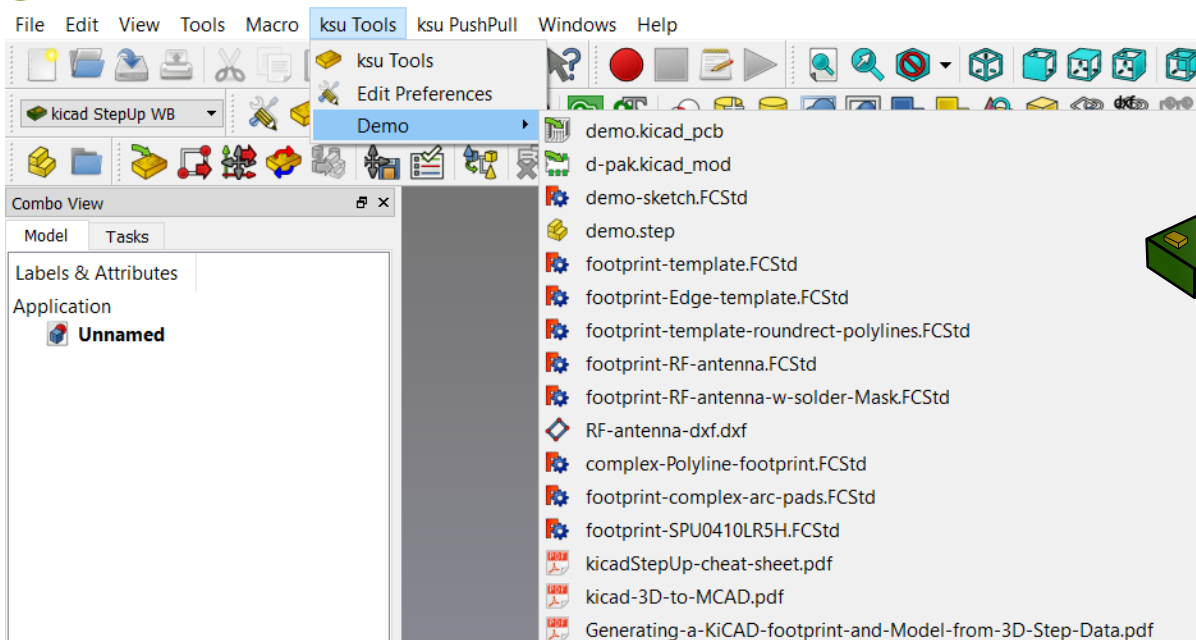
- **Manipulator workbench** useful to align and move assemblies and STEP models
Aligner Mover and Caliper are companions in 3D modelling

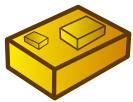


Defeaturing workbench useful for editing STEP models, removing some features from the model; defeaturing and repairing tools.



FreeCAD 0.18





KiCad StepUp tools cheat sheet

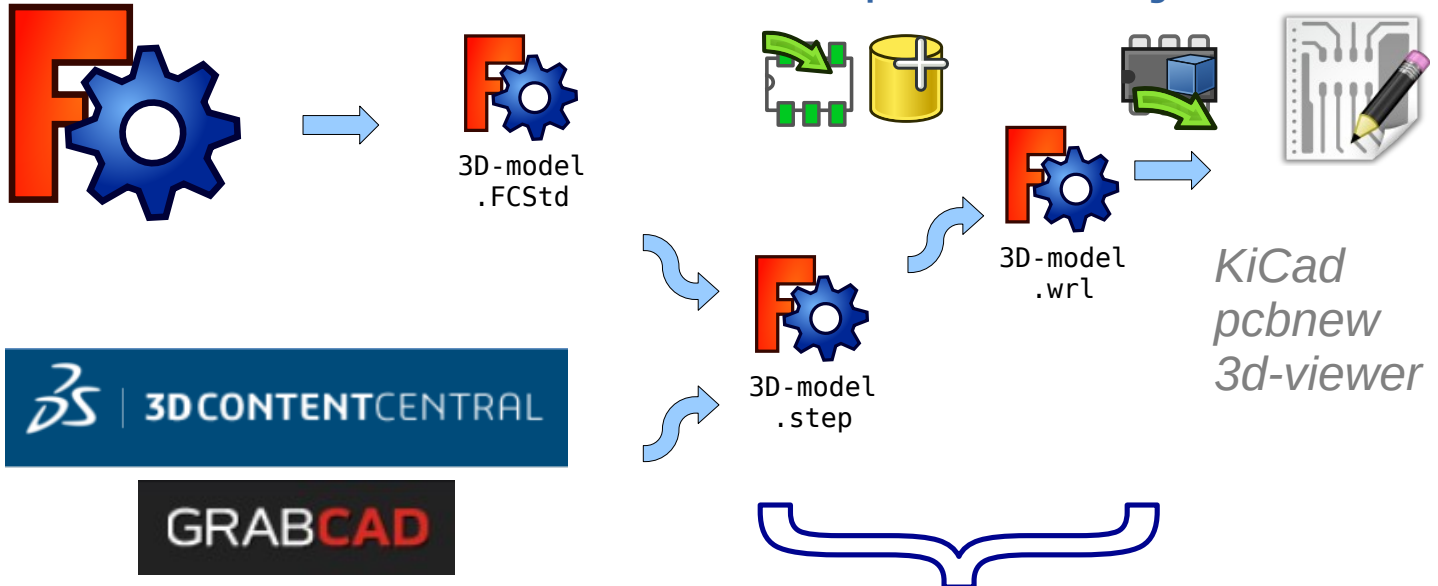
<https://github.com/easyw/kicadStepUpMod>

STEP compressed ['.stpZ']
VRML compressed ['.wrz']
formats are allowed

StepUp WorkFlow for 3D models

How to create a 3D model library for KiCad with StepUp tools

Use StepUp tools to Load the Footprint in FC and Align it



Use FreeCAD or any MCAD sw as 3D designer for a 3D model, or just download a 3D STEP model from on-line libraries

Place the exported models to the KISYS3DMOD folder

Use Manipulator WB

to align the STEP model to footprint
[Manipulator workbench](#)



Note:

when aligning a 3D model to a kicad footprint, StepUp takes care of:

- 2D footprint rotation of kicad for the footprint alignment
- vrml model z rotation

It is mandatory that the footprint has:

- x and y of the 3D model rotation set to 0
- x, y and z of the 3D model translation set to 0

The user has to check/modify, if needed, the part of 3D vrml/step model in kicad as following

```
(model path/name.wr1
(at (xyz 0 0 0))
(scale (xyz 1 1 1))
(rotate (xyz 0 0 0))
```

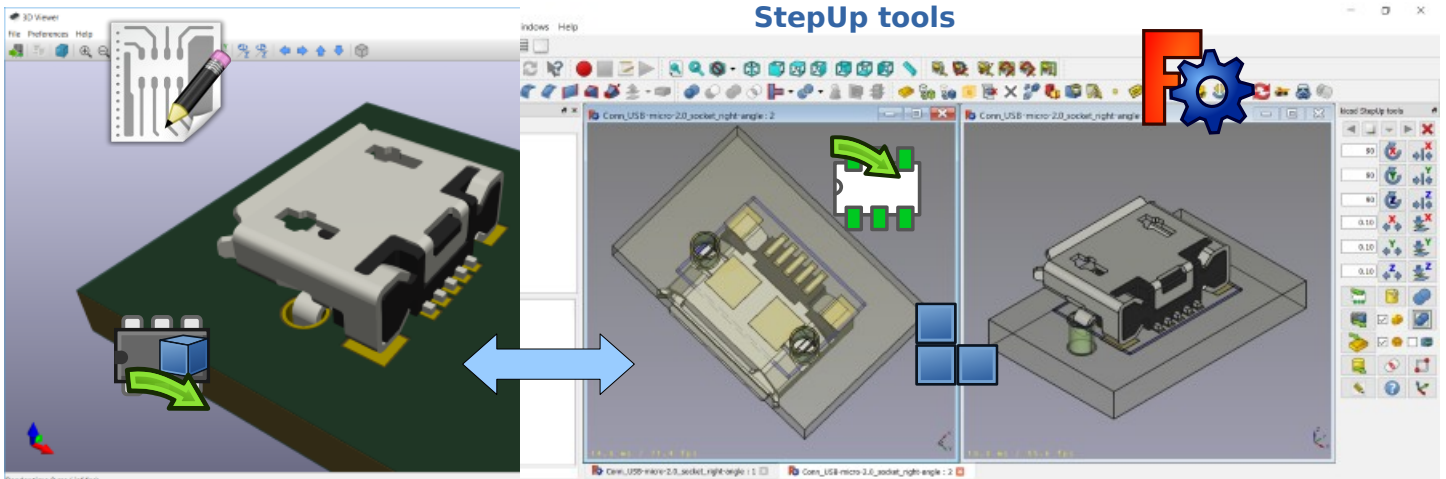
at (xyz 0 0 0) is mandatory, as much as scale (1 1 1)
rotate (xyz 0 0 z_value) can have a z rotation value

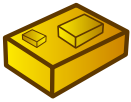
Video Tutorials

[Align Parts to Kicad footprint](#)

[Installing, Import 3D model, Exporting the Board](#)

StepUp tools





KiCad StepUp tools cheat sheet

<https://github.com/easyw/kicadStepUpMod>

Generating smaller 3D model with bounding boxes

Sometimes the need would be just a 3D MCAD model for analysis or simple space constraints, so a nice detailed component models in MCAD system may be not required or desired;

in that case it is possible to configure the exporter to:

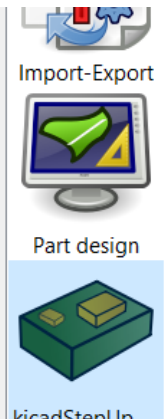
- skip 3D models by name
- skip models with a volume less than an assigned value
- skip models with a height less than an assigned value
- skip DNP models (add DNP to blacklist)

And then convert the remaining parts, or all but edge connectors, to bounding boxes

The result 3D MCAD model will have the accuracy of the pcb and assemblies only when needed, maintaining the model light as required.

Configuration file: Blacklist & BoundingBox parameters

Preferences Page



Bounding Boxes

Black List

Bounding Box LIST:

put here a list of 3D models to be converted to Bounding Box, separated by a comma

Examples:

ALL- -> all models will be converted to bounding boxes

R_0603, C_0603 -> these two models will be converted to BBox

LIST DSUB-15-HD_FH, DSUB-9_FH -> these two models will NOT be converted to BBox

☐ Make a

☐ Directly

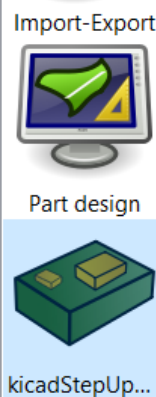
STEP exp

3D Loading

☒ Assem

☒ Enable

☐ Start Turn



Black List

Black List:

put here your model names that you don't want to load (e.g. smallest ones) separated by a comma.

STEP volume and height are also configurable.

3D Lc (volume=1 means all models with a volume < 1mm3 will not be included)

☒ A (height=1 means all models with a height < 1mm will not be included)

☒ E

☐ S

Examples:

r_0603,r_0402,c_0402,c_0603

height=1.0

volume=1.0

An empty list means all the models will be parsed.

White List

White List:

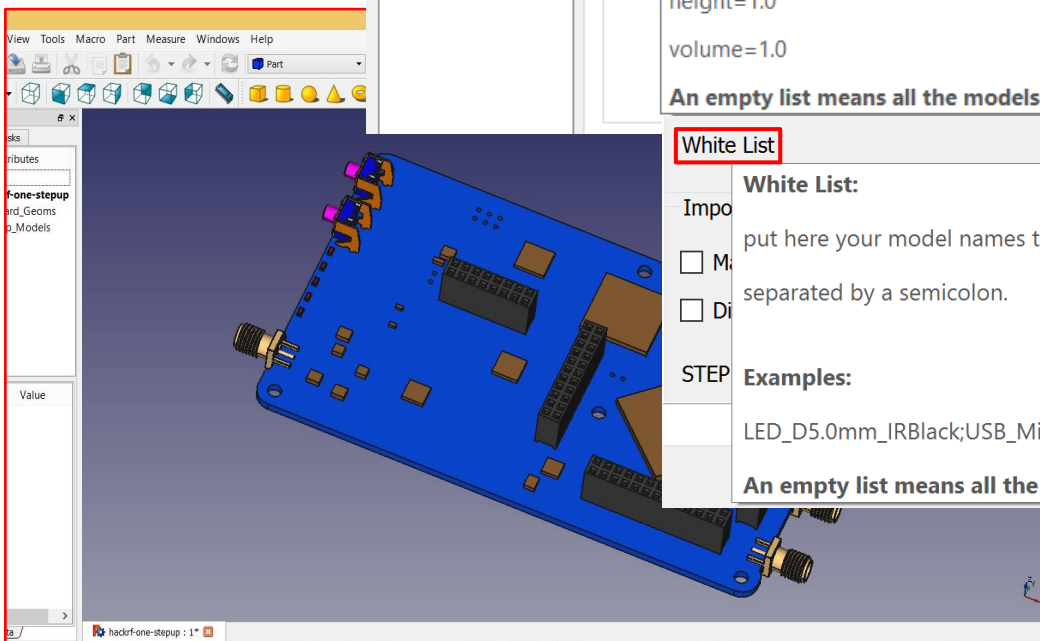
put here your model names that you must load (e.g. smallest ones) separated by a semicolon.

Examples:

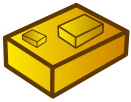
LED_D5.0mm_IRBlack;USB_Mini-B_Lumberg_2486_01_Horizontal

An empty list means all the models will behave as usual.

**KiCad StepUp:
Using
bounding boxes
for all but
connectors and
skipping
small parts**



**WhiteList:
for must load
models**



KiCad StepUp tools cheat sheet

<https://github.com/easyw/kicadStepUpMod>

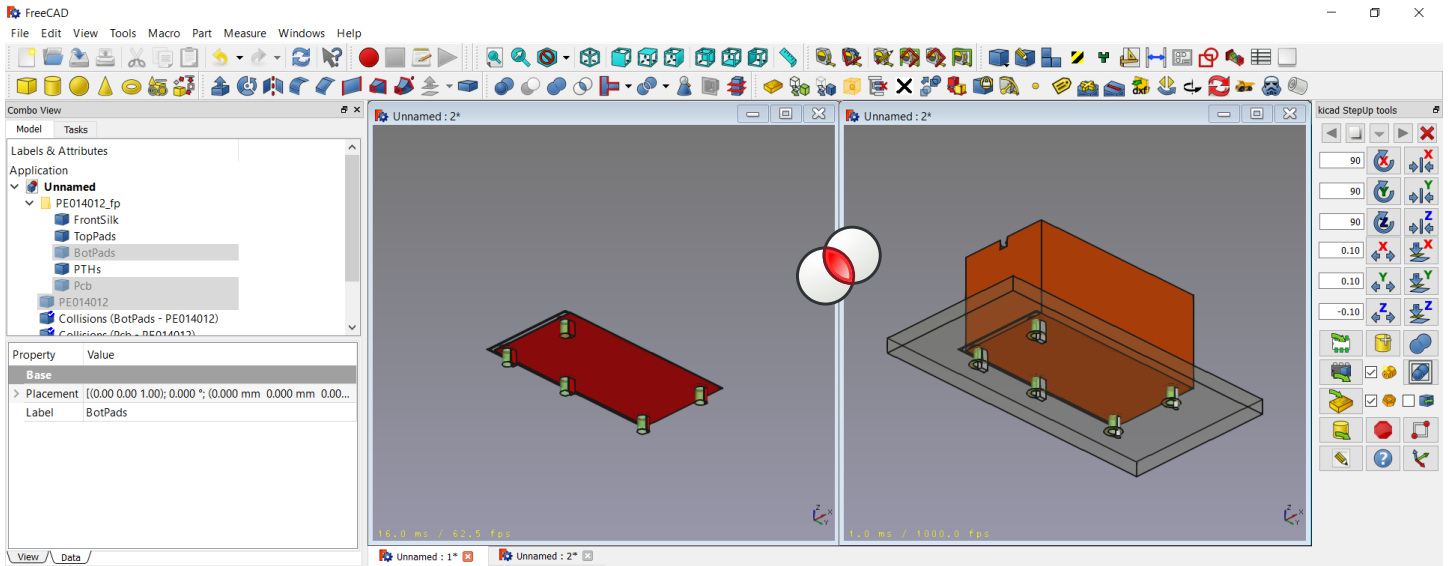
Check for Interference and mechanical constrains

With kicad-SteUp-tools it is also possible to detect collisions and check mechanical constrains:

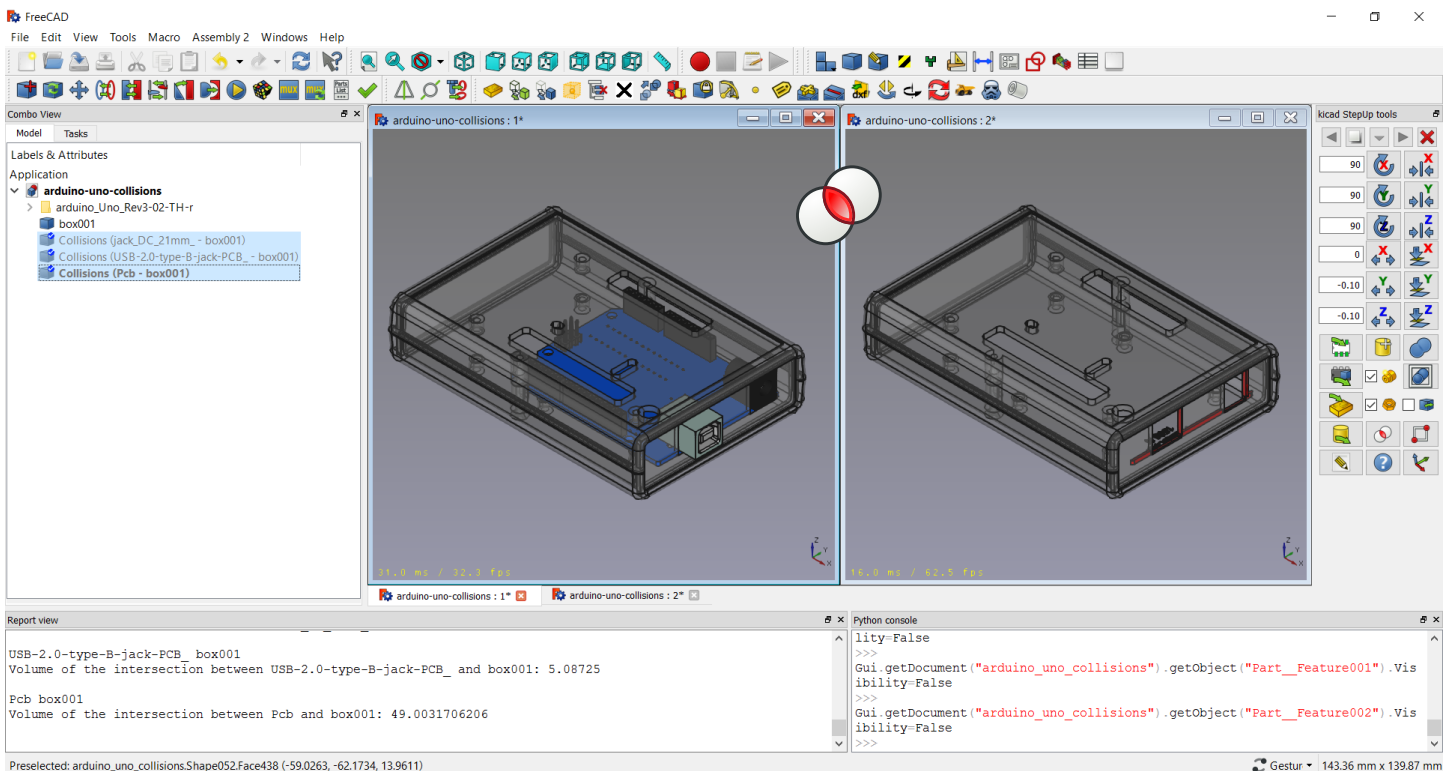
- detect collisions among part pins and drills for footprints
- detect collisions for enclosure clearance (between pcb with parts/connectors and enclosure)

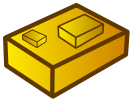


Interference checking for Footprints



Interference checking for PCB & Enclosure





KiCad StepUp tools cheat sheet

<https://github.com/easyw/kicadStepUpMod>

StepUp: ECAD MCAD Synchronization

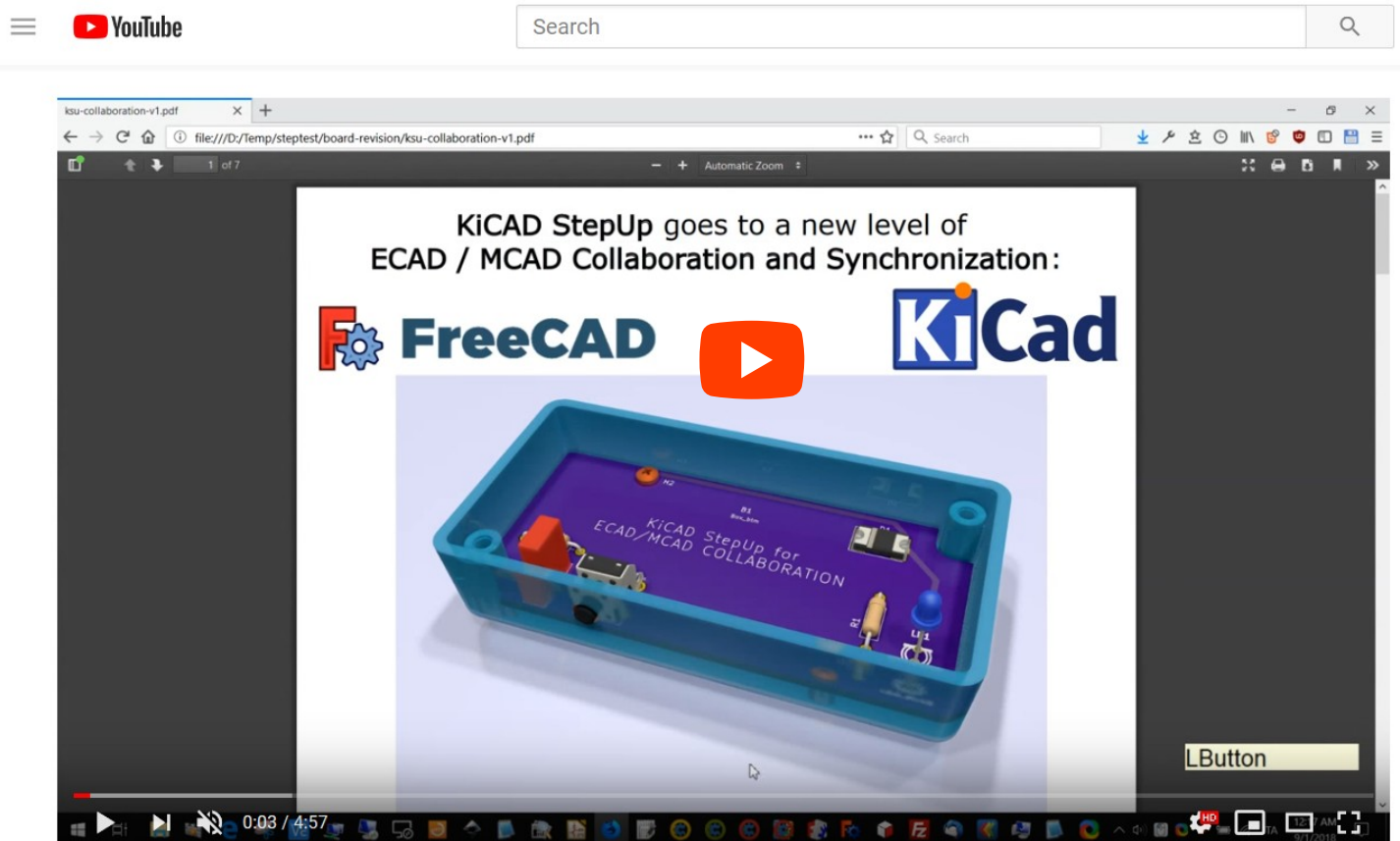
KiCAD StepUp goes to a new level of **ECAD / MCAD Collaboration and Synchronization**:

Push/Pull 3D model placement from/to KiCAD board to/from FreeCAD mechanical design.

It is possible to move 3D packages around on the 3D PCB mechanical sw, via both the X and Y axis.

The syncing process can be done even if the board is (fully) routed (i.e. when a new release requires some mechanical reviews).

ECAD MCAD integration is now fully implemented.



kicad StepUp: ECAD MCAD Synchronization



The ECAD MCAD
collaboration tutorial
ECAD MCAD Synchronization



Tips

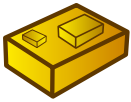
It is suggested to configure the preferences Page to use **grid origin** and **place a grid origin** to **kicad_pcb** file

PCB Placement **Grid Origin** ▼

KiCad StepUp Workbench



STEP compressed ['.stpZ']
VRML compressed ['.wrz']
formats are allowed



KiCad StepUp tools cheat sheet

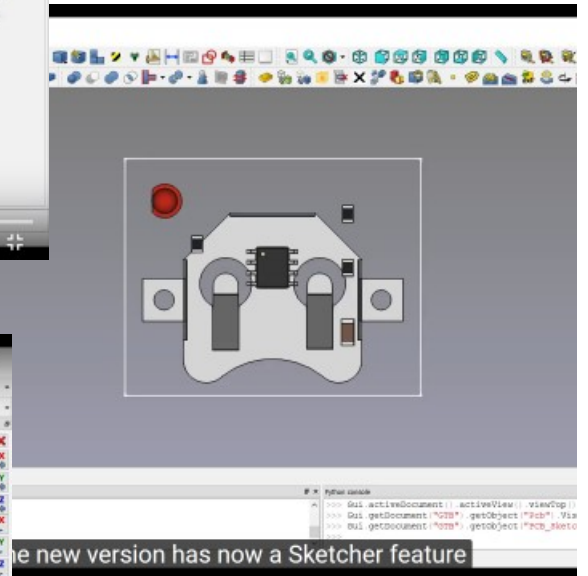
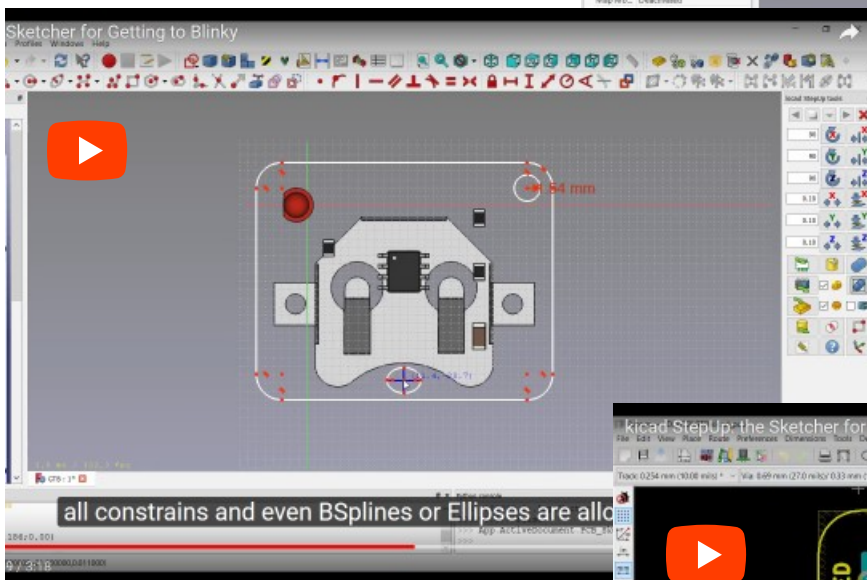
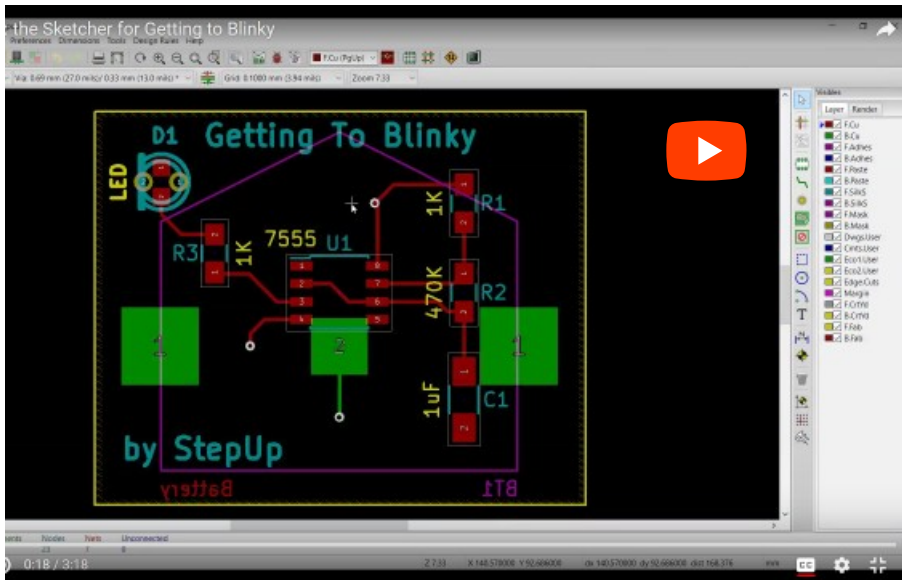
<https://github.com/easyw/kicadStepUpMod>

StepUp: The Sketcher

With kicad-SteUp-tools it is also possible to use FreeCAD Sketcher to create or modify a PCB Edge.

- create a new PCB Edge in FreeCAD Sketcher and PUSH it to kicad_pcb file
- read a PCB Edge from an existing kicad_pcb file and PULL it to FreeCAD Sketcher
- modify a PCB Edge in FreeCAD Sketcher and PUSH it to KiCad Board

Line, Circles, Arcs are supported and also **Bsplines or Ellipses** are supported and converted to KiCad compatible format



The Sketcher tutorial

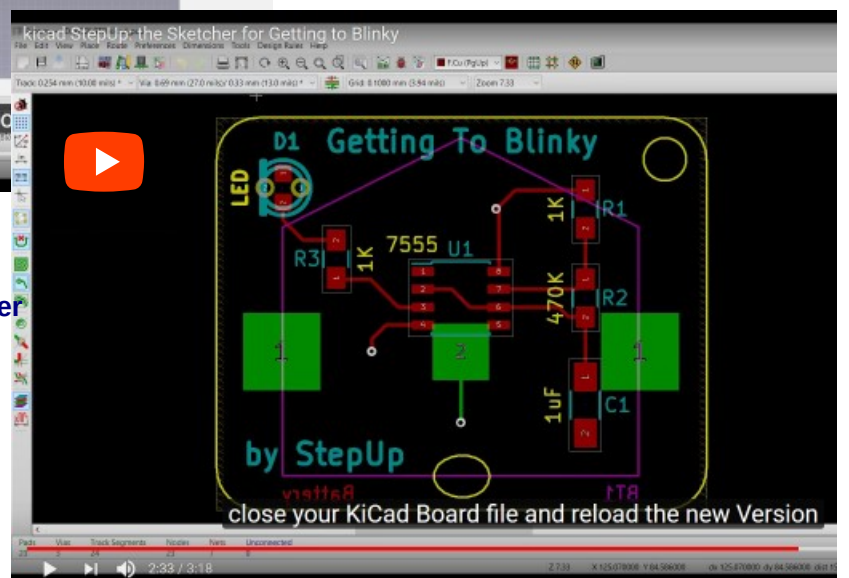
PUSH & PULL a PCB Edge using FC Sketcher

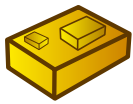
Tips

It is suggested to configure the preferences Page to use **grid origin** and **place a grid origin** to kicad_pcb file

PCB Placement

Grid Origin





KiCad StepUp tools cheat sheet

<https://github.com/easyw/kicadStepUpMod>

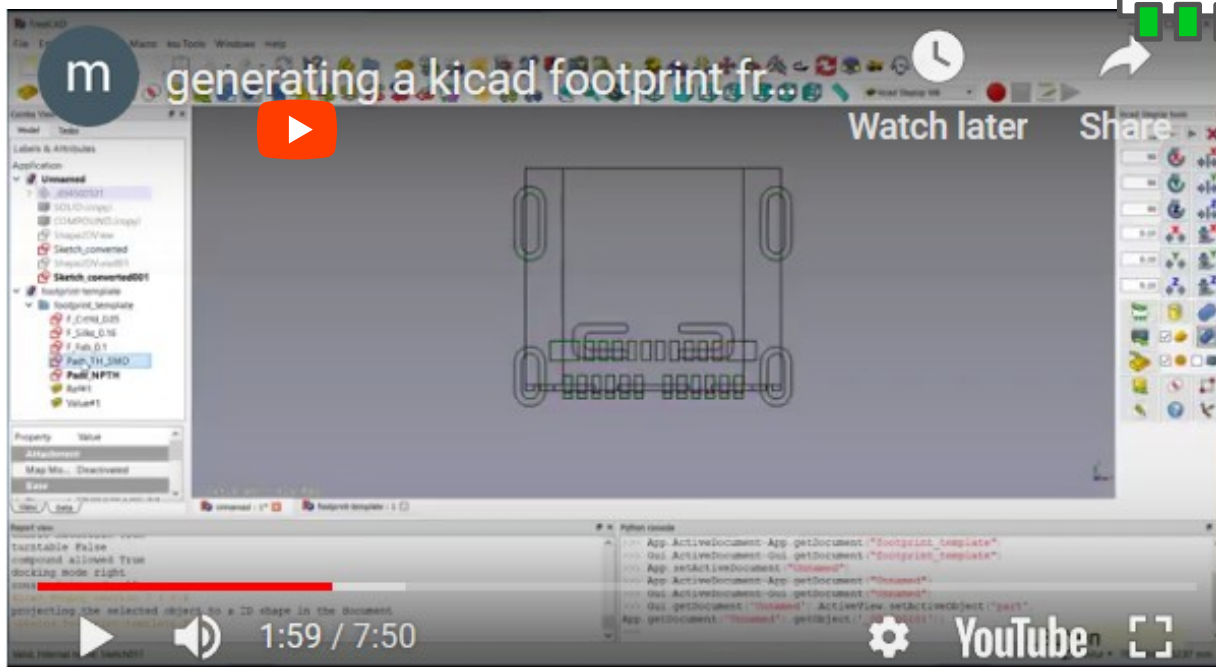
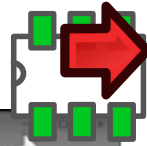
StepUp: The Sketcher for footprint generation

With kicad-SteUp-tools it is also possible to use FreeCAD Sketcher to create or modify a Kicad footprint.

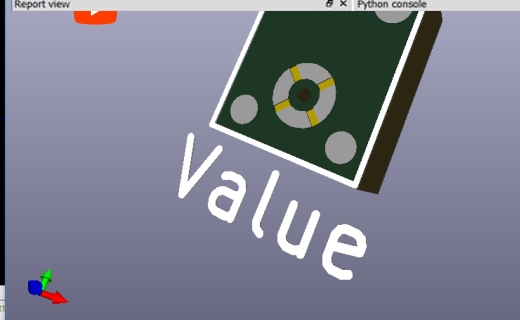
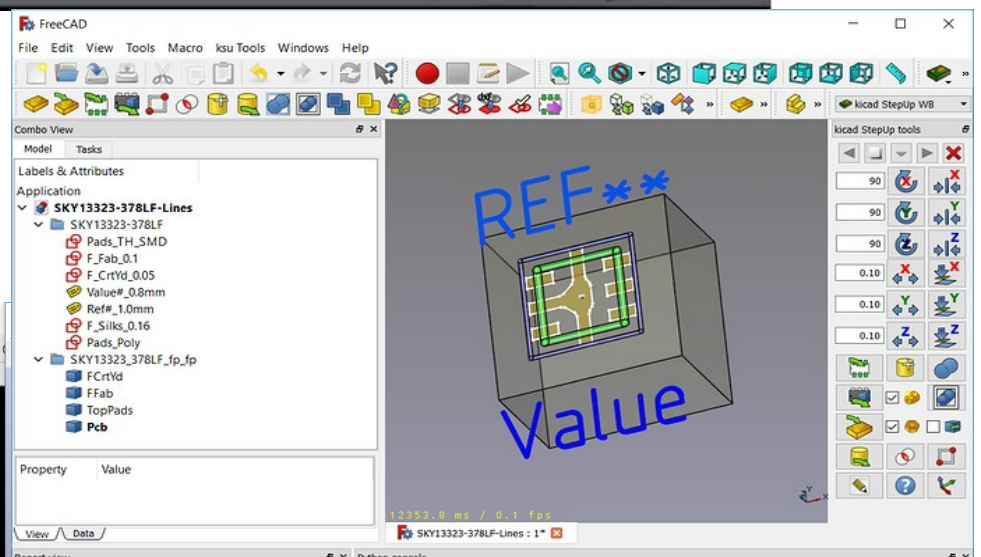
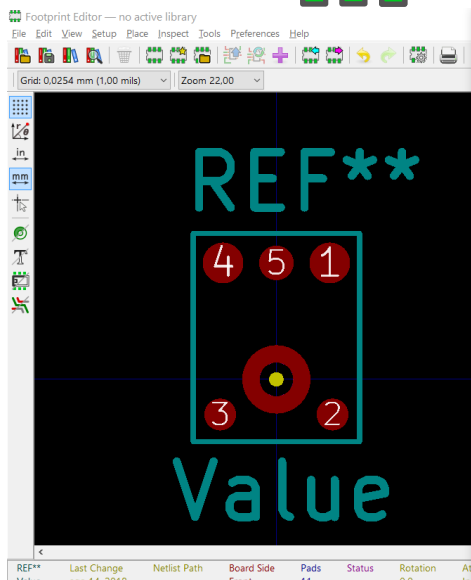
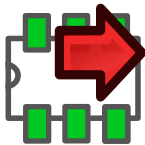
- create a new footprint in FreeCAD Sketcher and PUSH it to kicad_mod file
- modify an existing kicad footprint in FreeCAD Sketcher and PUSH it back to 'kicad_mod'

Line, Circles, Arcs are supported and also **Bsplines or Ellipses** are supported and converted to KiCad compatible format

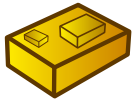
generating a kicad footprint from a 3D STEP model



The Sketcher for footprint: Tutorial



The Sketcher for footprint
@ **KiCad**INFO



KiCad StepUp tools cheat sheet

<https://github.com/easyw/kicadStepUpMod>

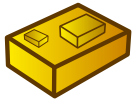
Preferences Page for configuring main parameters

All buttons have Tooltips

The screenshot shows the 'Preferences' dialog box with the 'General' tab selected. The left sidebar contains icons for 'General' (wrench and screwdriver), 'Display' (monitor), 'Import-Export' (document with 'F' and gear), 'Start' (blue arrow), and 'kicadStepUpGui' (3D component). The main area is titled 'General kSU 'Help Tips'' and contains several sections of settings:

- 3D Prefix working folder**
 - main 3D folder location 'SYS3DMOD': /usr/share/kicad/modules/packages3d
 - secondary 3D folder loc 'ALT3DMOD': /home/mau/Downloads/kicad-packages3D-5.1.5
- PCB settings**
 - PCB color: Green
 - PCB Placement: Grid Origin
 - PCB Sketch mode (constraints): Coincident
 - minimum edge tolerance: 0.01
 - ☒ Virtual 3D Models loading
 - minimum Drill size applied to PCB: 0.0
 - Bounding Boxes: (empty text box)
 - Black List: (empty text box)
- Import/Export settings**
 - ☐ Make a Union of 3D models
 - ☐ Directly Export STEP after Loading
 - STEP export mode: Hierarchy
 - 3D Loading mode: Hierarchy
 - ☒ Assembly3 Links allowed
 - ☐ Assembly3 LinkGroups allowed
 - ☒ Enable Materials for VRML exporting
 - ☐ Enable compressed VRML file ('.wrz') generation exporting
 - ☐ Enable compressed STEP file ('.stpZ') generation exporting
 - ☒ Display STEP import settings warning
 - ☐ Start Turntable after loading
- Mechanical Check**
 - ☒ Generate Sketches for differences on PCB Edge

At the bottom of the dialog are buttons for 'Help', 'Reset', 'Apply', 'Cancel', and 'OK'.

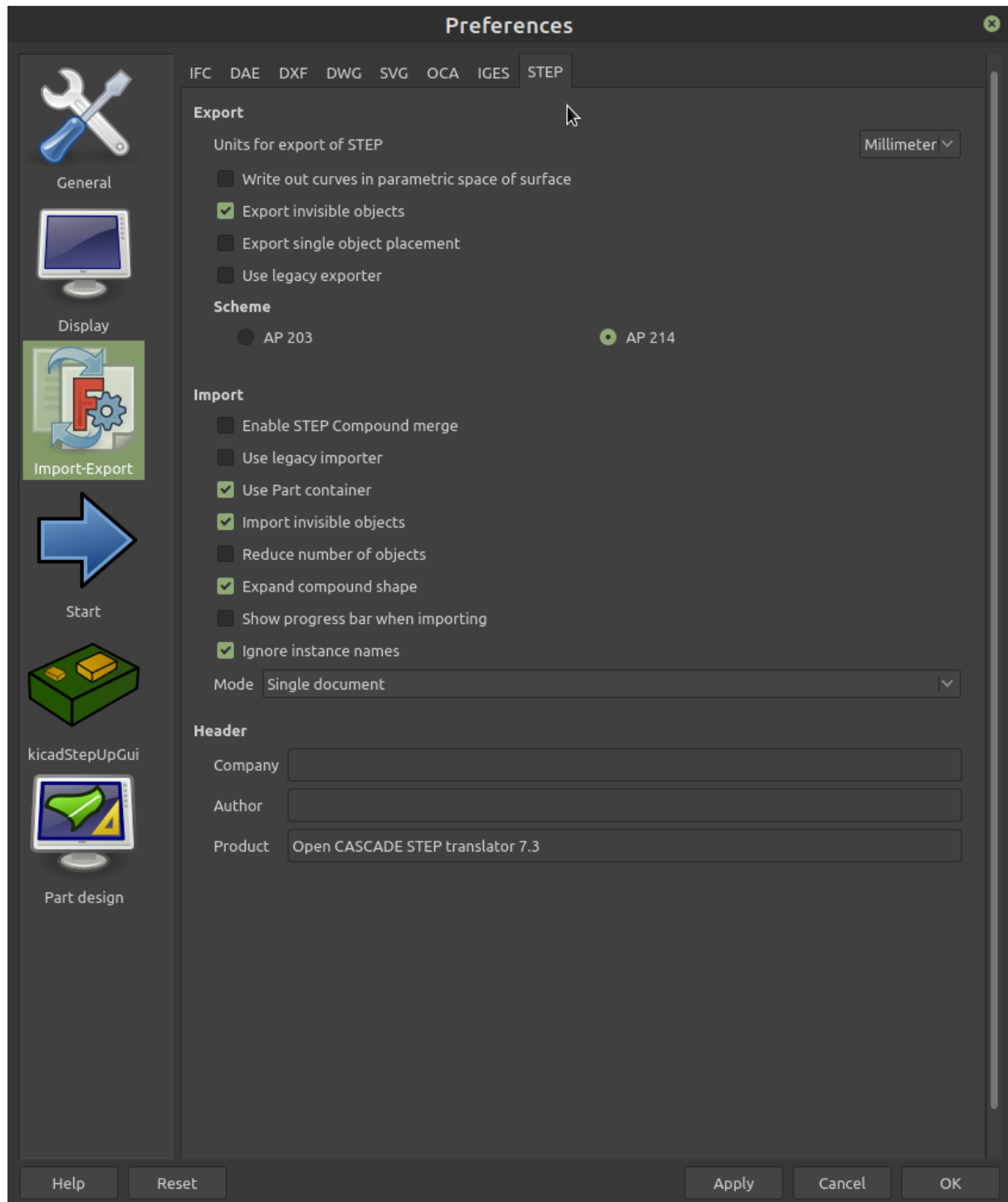


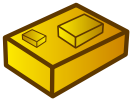
KiCad StepUp tools cheat sheet

<https://github.com/easyw/kicadStepUpMod>

Preferences Page for configuring STEP Import Export (FC0.19)

All buttons have Tooltips



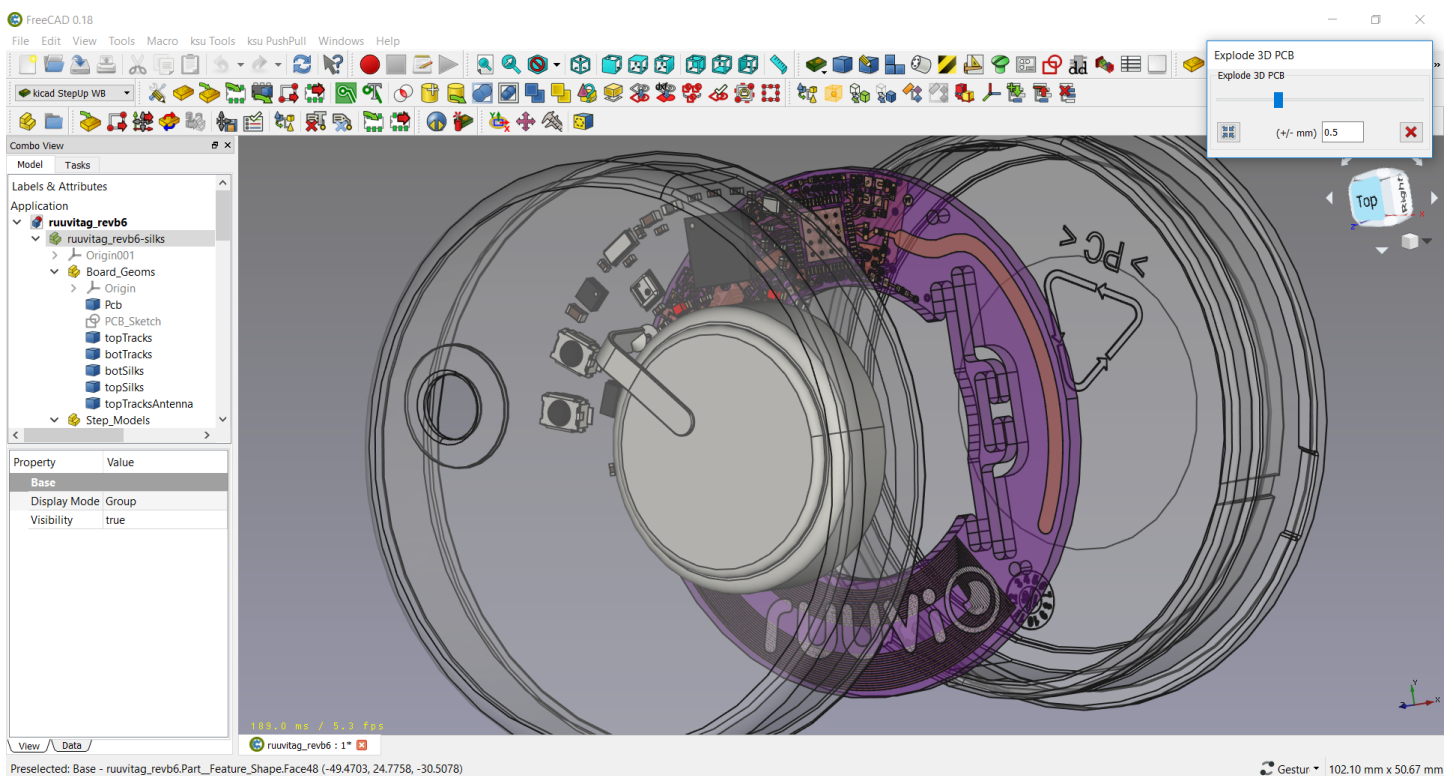
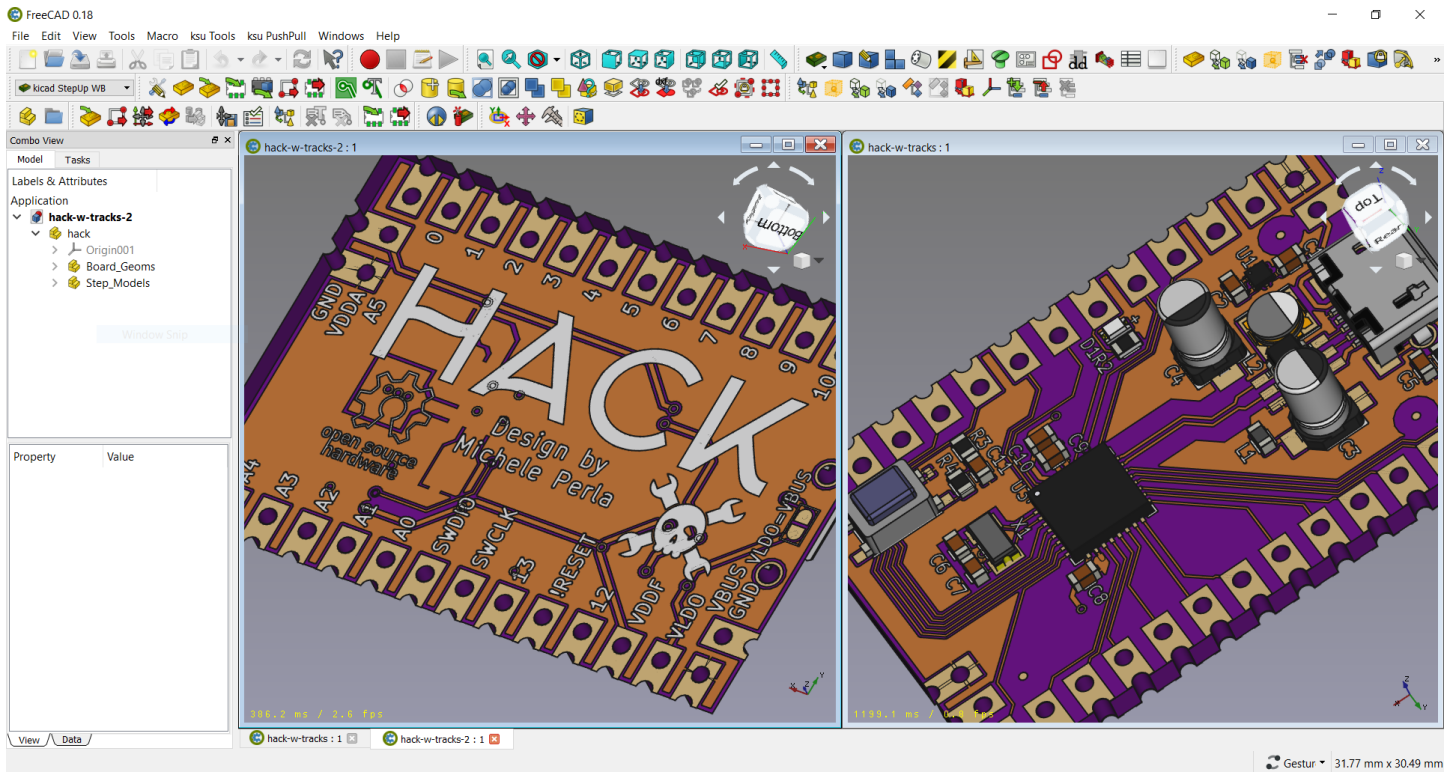


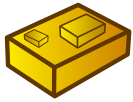
KiCad StepUp tools cheat sheet

<https://github.com/easyw/kicadStepUpMod>

Tracks and SilkScreen MCAD integration

New ability to import Top and Bottom tracks and SilkScreen layers





KiCad StepUp tools cheat sheet

<https://github.com/easyw/kicadStepUpMod>

Tracks MCAD integration

New ability to import Top and Bottom tracks and SilkScreen layers
(for PCB signal integrity and EMC simulation design please refer to [fcad_pcb](#) repo)



Top and Bottom tracks are imported directly from 'kicad_pcb' source file into FreeCAD designing document.



Top and Bottom SilkScreens are imported from Top and Bottom DXF files, exported from KiCAD source file.

KiCAD export configuration

Plot

Plot format: DXF Output directory:

Included Layers

- ☐ F.Cu
- ☐ B.Cu
- ☐ F.Adhes
- ☐ B.Adhes
- ☐ F.Paste
- ☐ B.Paste
- ☒ F.Silks
- ☒ B.Silks
- ☐ F.Mask
- ☐ B.Mask
- ☐ Dwgs.User
- ☐ Cmts.User
- ☐ Eco1.User
- ☐ Eco2.User

General Options

- ☐ Plot border and title
- ☐ Plot footprint values
- ☒ Plot footprint references
- ☐ Force plotting of invisible values / refs
- ☒ Exclude PCB edge layer from other layers
- ☒ Exclude pads from silkscreen
- ☐ Do not tent vias
- ☐ Use auxiliary axis as origin

DXF Options

- ☒ Plot graphic items using their contours
- ☒ Use Pcbnew font to plot texts

Drill marks: None

Scaling: 1:1

Plot mode: Filled

Default line width: 0.15 mm

☐ Mirrored plot

☐ Negative plot

☒ Check zone fills before plotting

Export units: Millimeters

Output Messages

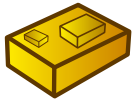
Show: ☐ All ☒ Errors ☒ Warnings ☒ Actions ☒ Infos

Run DRC...

Plot

Close

Generate Drill Files...



KiCad StepUp tools cheat sheet ^h

<https://github.com/easyw/kicadStepUpMod>

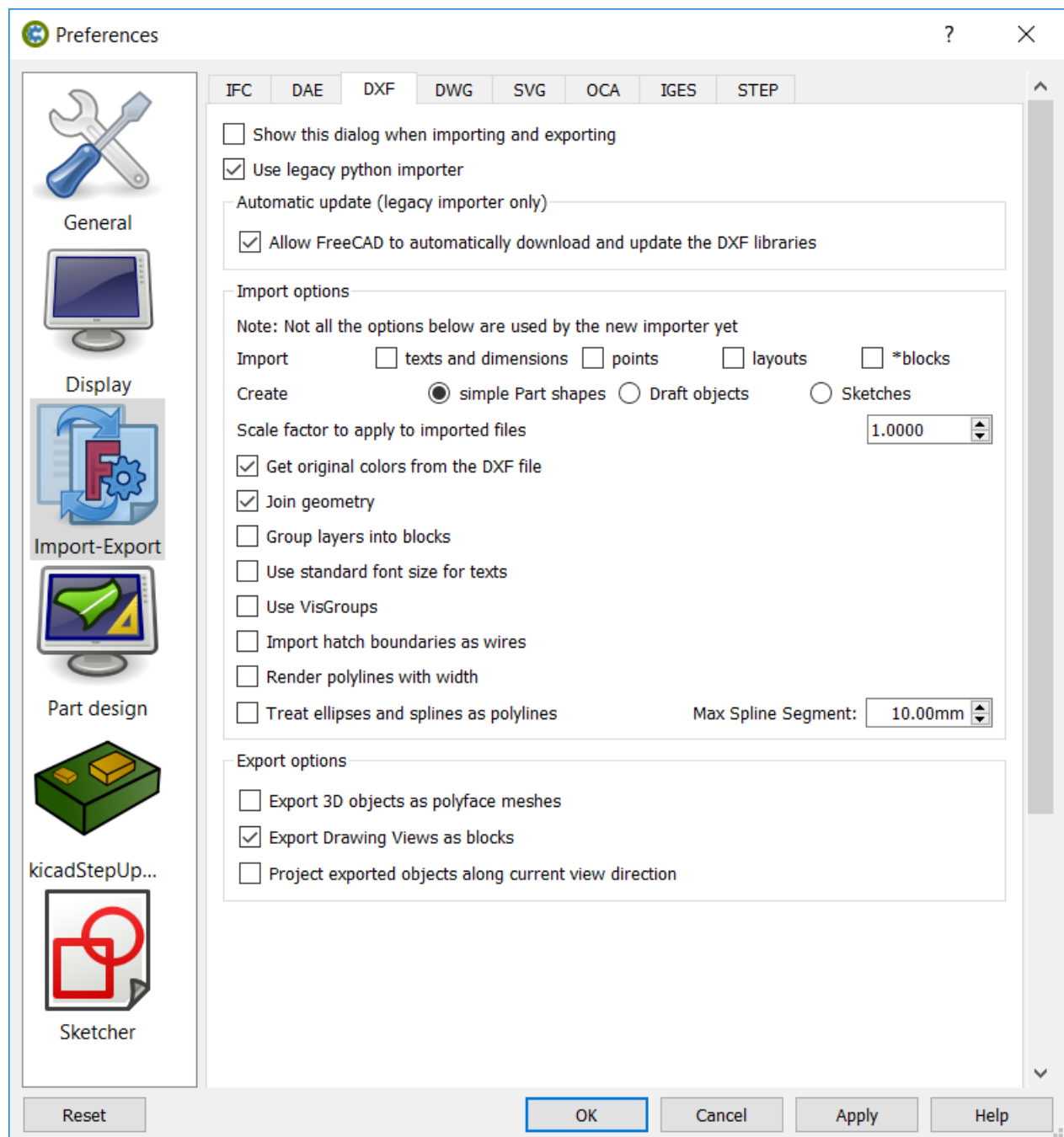
Tracks MCAD integration

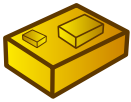
New ability to import Top and Bottom tracks and SilkScreen layers



Top and Bottom SilkScreens are imported from Top and Bottom DXF files, exported from KiCAD source file.

FreeCAD import configuration





KiCad StepUp tools cheat sheet

<https://github.com/easyw/kicadStepUpMod>

StepUp Credits

kicad StepUp tools author is Maurice <https://github.com/easyw/kicadStepUpMod>

IDF import for FreeCAD - Milos Koutny (milos.koutny@gmail.com)

CadQuery module - CadQuery FreeCAD module <https://github.com/jmwright/cadquery-freecad-module/>

hyOzd freecad macros - <https://bitbucket.org/hyOzd/freecad-macros>

FreeCAD-PCB - marmni <marmni@onet.eu26>

Kicad semantic parser - "Zheng, Lei" https://github.com/realthunder/fcad_pcb

Copyrights

this document and kicad StepUp tools are Copyrighted © 2015 2016 2017 by Maurice.

Kicad STEPUP™ is a TradeMark and cannot be freely useable.

This program is free software; you can redistribute it and/or modify it under the terms of the GNU Affero General Public License as published by the Free Software Foundation to ensure cooperation with the community in the case of network server software; for detail see the LICENCE text file.

<http://www.gnu.org/licenses/agpl-3.0.en.html>

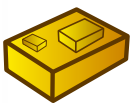
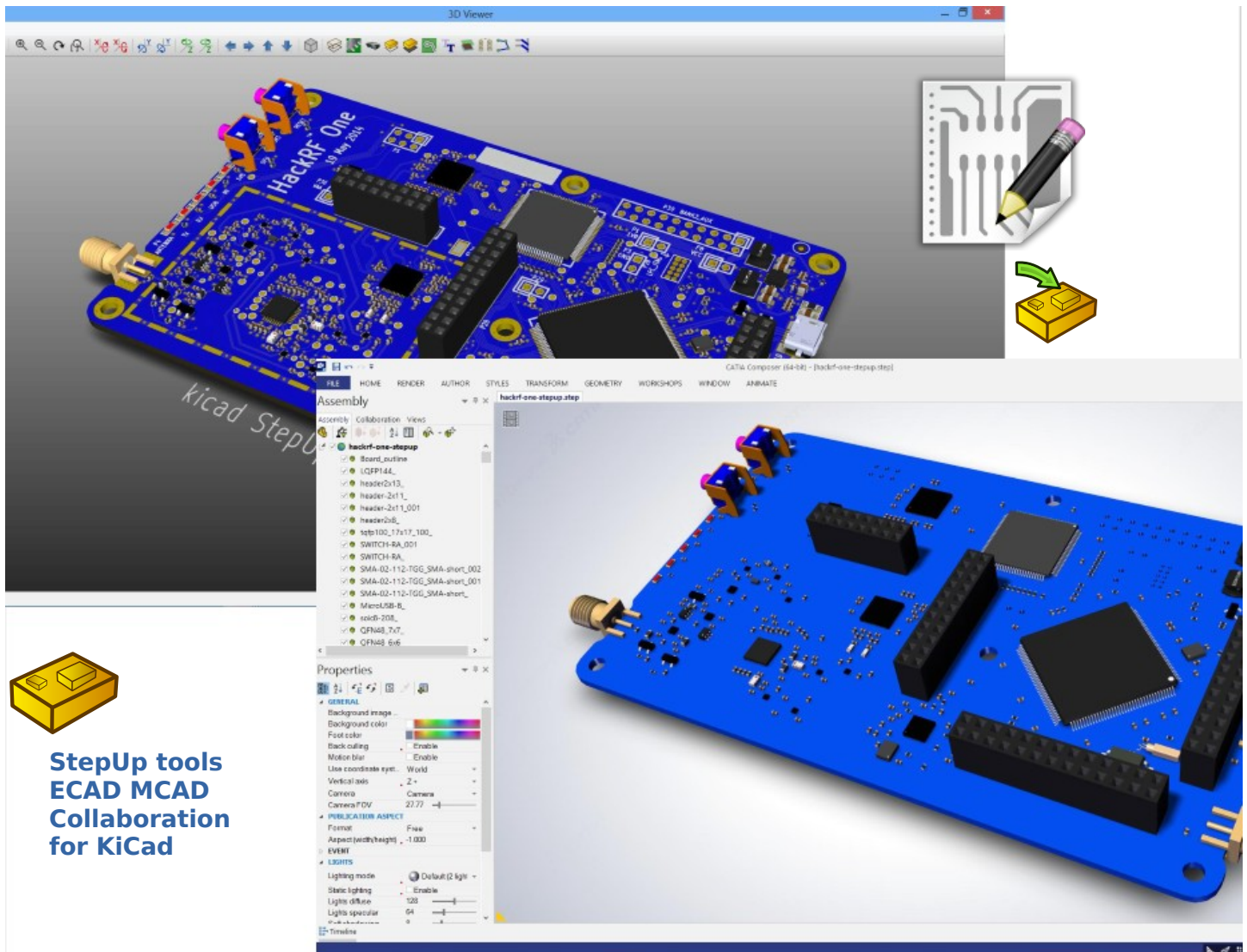
Moreover you have to include the original author copyright.

All trademarks within this guide belong to their legitimate owners.

Risk disclaimer

USE 3D CAD DATA AT YOUR OWN RISK

DO NOT RELY UPON ANY INFORMATION FOUND HERE WITHOUT INDEPENDENT VERIFICATION



**StepUp tools
ECAD MCAD
Collaboration
for KiCad**